Neural signatures of hyperdirect pathway activity in Parkinson's disease

Paper: Oswal et al. (2021). Human recordings, tractography and modelling.

DOI: 10.1038/s41467-021-25366-0

Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Network measurements and possible origins of beta synchronization. Helps distinguish frequency bands and pathways rather than treating beta as one homogeneous signal.

How does it answer the general questions?

Measured. STN/GPi recordings, MEG and tractography linked supplementary-motor-area–STN high-beta coherence (21–30 Hz) to hyperdirect fibre density. Directional analyses supported cortical drive at high beta; the corresponding relationship was absent for low beta (PDF pp. 1–7).

Mechanism. A coupled cortical/basal-ganglia model showed that enhanced high-beta hyperdirect input can provoke lower-beta subcortical synchronization. This is a model-based explanation supported by multimodal associations, not a selective clinical test proving that either band must be suppressed (PDF pp. 8–11).

Model relevance. Motivates adding cortical populations and hyperdirect coupling to an STN–GPe model. Transmission delays and time constants shape the simulated spectra. The full parameter set is referred to Supplementary Table 2; tract density and coherence should not be substituted directly for dynamical gains. The local main-article PDF alone does not supply a complete numerical notebook parameterization.

What papers does it cite that could provide more information?

- Nevado Holgado, Terry and Bogacz (2010), *Conditions for the generation of beta oscillations in the subthalamic nucleus–globus pallidus network*: delayed rate model (ref. 33, PDF p. 13).
- Hammond, Bergman and Brown (2007), *Pathological synchronization in Parkinson's disease: networks, models and treatments* (ref. 38, PDF p. 13).